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WHITEPAPER

The Digital-First Bank's Guide to AI in 2026





Thinking of implementing AI at your bank, digital bank, or lending institution? You wouldn't be alone. In just one year, financial institutions [invested more than \\$35 billion in AI](#), with that number predicted to reach a whopping \$97 billion by 2027. And it's no surprise this investment is being made; from hyper-personalised mobile banking experiences to automated credit decisions in seconds, AI-powered innovations are helping institutions serve customers in better, faster ways.

However, achieving these gains requires more than just well-designed algorithms; it demands a solid foundation of data and a strategy built for this new, AI-powered financial ecosystem. Because success with modern [AI in digital banking](#) boils down to a few essentials: reliable access to production-grade data, a way to work with that data off the live core system, and a consistent pipeline from raw data to AI-driven decisions.

Before rolling out chatbots or fraud detection models, banks must first ensure their data layer is robust and secure. If you've found yourself tasked with green-lighting an AI project at the digital banking division of your institution, modernising lending and deposits at a regulated bank, or building out a new AI layer in your fintech or MFI, this is the whitepaper for you. In it, we provide some easy-to-digest guidance for digital banks and financial institutions in dynamic markets, helping highlight some of the opportunities and challenges of AI and sharing how you can use AI to help your business grow. We explore the key AI applications transforming digital banking (from risk scoring and fraud prevention to chatbots and automation) and provide strategic guidance on implementing AI responsibly. We discuss technical considerations, from integrating AI with core banking systems to maintaining regulatory compliance in different jurisdictions. Throughout, we underscore how your data layer lays the groundwork for effective AI deployment. By the end of this paper, we hope you'll have a clear grasp not only on the promise of AI in banking, but also the practical steps to harness it in 2026 and beyond.

\$97b

will be invested in AI by financial institutions by 2027

Source: World Economic Forum (2025)



Why AI projects fail: it's your data, not your algorithm

Every AI initiative stands on the shoulders of data.

The quality and accessibility of data can make or break an AI project. If your institution cannot reliably extract and unify its own data, it isn't ready for AI.

However, for many dynamic market banks, data often resides in siloed systems or legacy cores that were not built with real-time analytics in mind. This leads to fragmented views of customers and operations, undermining any advanced analytics or machine learning effort. A robust, accessible data layer is the prerequisite to leverage AI effectively; no algorithm can compensate for unstable foundations.

AI readiness is a banking discipline, not just a data project. Early-majority institutions that succeed with AI tend to have:

- Governed change and clear audit trails in their core systems
- Strong financial accuracy and reconciliation across products and channels
- Stable digital and branch channels, with defined exception-handling paths
- Secure, controlled access to data for analytics and AI, with clear permissions
- Resilience and continuity plans if something fails
- Accountable delivery and support from their core and technology partners

So, what does a strong data foundation look like?

It means having a single source of truth for all your transactional and customer data, kept up-to-date and readily queryable without disrupting daily operations. Many banks achieve this by creating a secure replica of the core banking database, used exclusively for analytics and AI model development. This replica is refreshed in near real-time but isolated from the live core so that heavy queries or experiments don't risk any downtime on production systems.

For example, institutions running on Oradian get these foundations as part of the core banking platform: a cloud-native, bank-grade core that ensures financial accuracy, continuity, and multi-channel stability, and Database Access, which is a secure, read-only mirror of the production PostgreSQL database, continuously updated and typically stored on the bank's own infrastructure. With complete, full-fidelity data available off-core, teams can plug into BI tools or ML platforms and work with the full spectrum of data, including historical transactions, user behaviour logs, and metadata, that standard reports or limited APIs might not expose. This combination of core and data layer is AI-ready by design, as it unlocks the raw ingredients that AI models need to truly perform.



The unglamorous work that truly makes AI possible

In practice, ensuring data readiness involves several steps. First, banks must integrate their various data sources into a unified repository or data warehouse; that includes their core banking system, mobile apps, and payment channels. They should establish basic pipelines to move data from the core system into an analytics environment reliably and repeatedly. It's important to define common data definitions and to clean up obvious quality issues before any modeling; that means no duplicate records and consistent customer IDs. It's also essential to get clarity on which teams can access what data, and safeguards to maintain privacy and security. A stable, repeatable process that gets the right data in front of the right people is far more important than chasing a new AI tool.

Ultimately, investing in your data layer is the first and most important AI investment. Banks in emerging markets should treat data consolidation and accessibility as strategic priorities on the road to AI. This foundational work lays the groundwork for all the AI applications discussed in the next section. With clean, comprehensive data at your fingertips, you can move confidently into deploying AI for risk scoring, fraud detection, customer service, and more. One proven way of achieving this

is with a live, governed replica of your core database. Without a solid data layer, even the most powerful AI algorithms will be operating blind or on shaky ground.



Where AI actually works in banking right now

Dynamic markets like those in Nigeria, the Philippines, and Indonesia present fertile ground for high mobile adoption and underserved customer segments, while evolving financial ecosystems create unique opportunities for AI to make an impact.

Here are some of the key AI use cases that are currently transforming banking in these regions.

How AI finds creditworthy borrowers in your data

[Fintech Pulse](#) reports that millions of consumers and small businesses in emerging markets have little or no formal credit history, making their credit difficult to verify according to conventional scoring models. AI changes the game by analysing alternative data sources and behavioural patterns to assess creditworthiness in new ways.

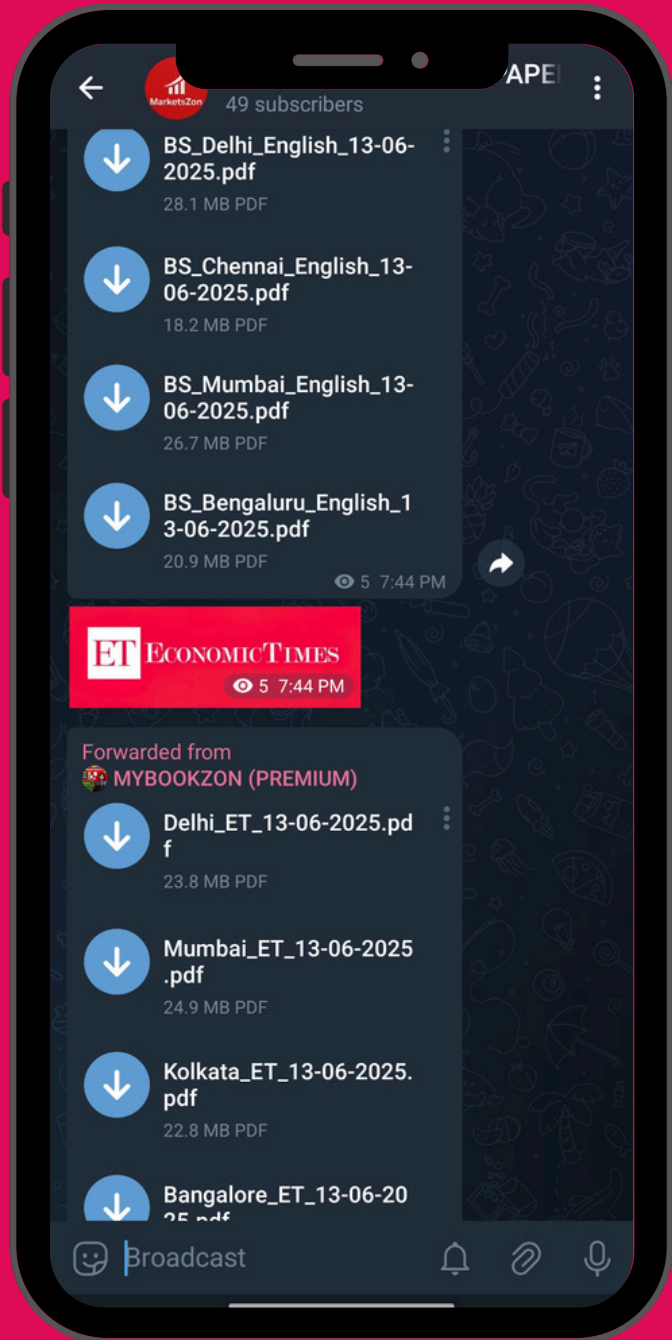
For example, machine learning models can ingest data from sources like telecom usage, mobile wallet transactions, utility payments, e-commerce activity, and even social indicators to paint a richer picture of a borrower's financial reliability. [Fintech Pulse](#) goes on to highlight that markers such as paying phone bills on time or maintaining a stable device location can be correlated with good credit behaviour, while irregularities might flag higher risk. These complex, multi-dimensional signals would overwhelm traditional manual underwriting, but AI excels at finding predictive relationships in such raw and fragmented data.

The result is more accurate and inclusive credit scoring. AI-driven scoring models can update borrower risk profiles in real-time, for instance, immediately incorporating the latest mobile payment transactions or repayment behaviors, which makes lending decisions faster and more scalable even in markets with limited financial infrastructure. Studies have found that AI credit scoring systems improve default prediction accuracy by [15–25% compared to legacy methods](#), enabling lenders to approve more loans to qualified borrowers while reducing default rates by as much as 30%. These gains come from analysing a broader dataset than a human underwriter could and dynamically adjusting risk models as new data streams in. Just as importantly, AI-based credit [assessments expand financial inclusion](#): small entrepreneurs, gig workers, and individuals with thin files can now get credit because alternative data sets like consistent mobile wallet deposits or on-time rent payments may now be able to prove their creditworthiness.

The use of data like this is quickly becoming the norm in developing markets, with alternative data enabling the assessment of [1.7 billion unbanked](#) individuals across the globe and signals like those coming from phone usage and bill payments supporting the credit assessment of millions more.

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Lenders benefit too by being able to uncover new customer segments while being alerted to the early warning signals of risk that traditional models might miss, for instance, a sudden drop in mobile top-up frequency might precede loan delinquency.

To fully leverage AI for risk scoring, institutions must ensure data diversity and quality. This means tapping into telecom data, transaction histories, and other alternative sources via partnerships or APIs, and then feeding that data into AI models through a governed pipeline.

Banks should, however, also be mindful of ethical considerations: AI models can inadvertently pick up biases present in training data, such as socio-economic or demographic biases. For that reason, it's essential to manage and monitor these models for fairness and to maintain transparency about how decisions are made. AI-enabled credit decisions should be explainable and responsibly governed.

When implemented thoughtfully, AI risk scoring can significantly improve portfolio quality, speed up loan approvals and bring a vast new population into the fold of finance, all of which can provide a transformative impact for emerging markets.

Catching fraudsters before they clean you out

As digital banking transactions surge, fraud and financial crime also become more sophisticated. Fortunately, so have AI-driven defences, with 90% of financial institutions using it for fraud detection and AI systems achieving a [90-99% accuracy rate](#) as opposed to the previous 35-70% accuracy rate of traditional models.

Fraud detection was one of the earliest arenas for AI in finance and it remains a top priority for banks given the high stakes of preventing breaches, scams, and money laundering. Sources show that [AI systems can analyse](#) real-time transaction patterns, user behaviors, and device fingerprints across millions of events to spot anomalies that might indicate fraud, all faster and with more accuracy than traditional rule-based systems. For example, if a normally low-balance savings account in Nairobi suddenly attempts a large transfer from a new device at 2am, an AI model could instantly flag this as suspicious and either block the transaction or route it for human review. Machine learning models excel at identifying subtle patterns and correlations that would elude manual monitoring, such as combinations of location, timing, frequency, and amount. This means banks can catch fraudulent activities in real time, rather than after the damage is done.

The more historical fraud and legitimate transaction data you feed into these models, the better they become at distinguishing the signal from the noise. That's why having a complete, unified view of customer accounts, login activity, and transaction history is vital. By feeding a fraud engine a comprehensive view of transactions, accounts, devices, and customer history, banks dramatically improve their ability to detect anomalies and reduce false positives.



Test before you deploy

In practice, this could involve running machine learning models in shadow mode on live data by testing new fraud detection algorithms on a replica of current transactions to see how they perform without yet impacting customers. This shortens the loop between discovering a new fraud pattern and updating the system to respond to it, a critical capability as criminals constantly change tactics. With this method, AI can identify fraud with unprecedented speed and precision, potentially preventing losses that would otherwise occur in the gaps of human oversight.

A safe AI pilot for fraud typically includes:

- Running in shadow mode or parallel with existing rules for a period of time
- Clear success criteria agreed up front, for example, fewer losses at an acceptable false-positive rate
- Documented approvals and governance, including sign-off from risk and compliance
- Fallback and override paths, so any blocked transaction can be escalated and decisions can revert to existing processes if needed

Framing pilots this way reassures risk-conscious stakeholders that AI is being introduced as a controlled enhancement to existing defences, not a risky replacement.



Casestudy

How M-Pesa already does this



We're already seeing success stories in the market. For instance, Kenya's [M-Pesa mobile money platform](#) uses predictive AI algorithms to spot and stop fraudulent transactions in real time, helping to maintain trust among its millions of users. The system watches for anomalies in user behavior and transaction flows, and can automatically halt suspicious activities or alert investigators.

From verifying identities to providing personalised fraud warnings, AI tools are also being used to assist customers and front-line staff in security. Chatbots, for example, can notify customers of unusual account activity in plain language or guide them through security steps. All these efforts strengthen the institution's security posture and protect customers, a non-negotiable aspect of digital banking growth.

The practical side of AI fraud prevention

From an implementation standpoint, fraud detection AI must be integrated thoughtfully into banking operations. It should tie into transaction processing systems and case management workflows for investigators. Banks need a robust data pipeline to feed transaction data to AI models without lag.

As always, quality data is key: incomplete or delayed data feeds can lead to missed detections or false alarms. Additionally, transparency is crucial: if an AI flags a transaction or declines it, the bank should be able to explain why to a customer or regulator and allow for human override if needed. With these pieces in place, AI-driven fraud detection becomes an around-the-clock sentinel, scrutinising every digital interaction with a level of vigilance that manual systems simply can't match. For banks in emerging markets, this can mean being able to scale digital services securely with a foundation of trust for further innovation.



Get customer service without the wait times with AI chatbots

The rise of conversational AI is turning erstwhile chatbots into intelligent virtual agents that actually enhance the customer experience while reducing costs. In markets where mobile-first banking is common, AI-powered chatbots on web, app, or messaging platforms allow banks to offer 24/7 customer support and onboarding in a scalable way.

These are not the clunky bots of the past with limited scripted answers. Modern banking chatbots use advanced NLP and even gen AI to understand a wide range of customer queries and respond with relevant, context-aware answers. They can check balances, assist with fund transfers, answer FAQs about loans or fees, help reset passwords, and much more all in real time and often indistinguishable from a human interaction in quality.

For customers, this means instant service at any hour without waiting on hold or visiting a branch, which is particularly valuable in regions where branch access might be limited or call centers are overwhelmed. Indeed, banks deploying AI chatbots have reported tangible improvements: some saw a [30–40% increase in customer engagement](#) on digital platforms and a 25% reduction in drop-off rates once AI agents were handling inquiries promptly. More engagement and less frustration translate to higher customer satisfaction and retention.

AI chatbots also bring a new level of personalisation and proactivity to customer service; since they can tap into internal databases and past customer data, they can tailor their conversation to the individual. Micro-personalisation at scale has historically been difficult, but AI makes it possible to deliver the right message at the right time for thousands of customers simultaneously, leading to exceptional results. For instance, PwC reports that one institution achieved a [40% reduction in commercial banking verification costs](#) thanks to AI powered onboarding. They go on to share that embracing AI could help banks gain an efficiency ratio improvement of [15-percentage-points](#).

For banks and lenders, the benefits of AI customer service automation are twofold: better customer experience and significant operational savings. By handling the bulk of routine queries and tasks, AI agents free up human staff to focus on complex or high-value interactions like handling complaints, negotiating with delinquent borrowers, or advising on complex products. This augmented service model can reduce customer support costs substantially.

To implement chatbots successfully, institutions should integrate them with core banking and CRM systems so that your AI tool can fetch account details, execute transactions, and log interactions. Ensuring your tool is trained on a wide variety of customer phrases and languages prevalent in the market will improve its accuracy. It is also wise to have a fallback to human agents if the tool cannot resolve an issue or if it detects an emotionally upset customer.



Stop treating every customer the same with AI-powered personalisation at scale

Beyond automated service, AI enables a new frontier of personalised banking, which provides any digital bank with an essential edge given 80% of consumers say they are more likely to engage with an institution that offers personalised experience.

AI-driven personalisation uses analytics on customer data to segment customers into smaller micro-segments or, in some cases, to treat them each individually. This way, banks can move away from a one-size-fits-all approach to one where each customer feels the bank truly understands and anticipates their needs.

Examples of AI personalisation

Globally, banks leveraging AI have begun to unlock [hyper-personalisation at scale](#), which not only improves customer satisfaction but also drives revenue growth by matching customers with the right products at the right time. In fact, [77% of banking leaders](#) say that personalisation directly boosts customer retention and loyalty and that when people feel their financial provider is proactively looking out for them, they are less likely to switch to competitors. By implementing AI-powered personalisation, [ING experienced a 25% increase](#) in customer satisfaction, while Wells Fargo's predicted approach gained them a 15% increase in sales.

AI facilitates this by sifting through massive datasets to find patterns and triggers. It might identify, for instance, a group of customers who frequently use a certain money transfer service and then cross-offer them a new remittance product with lower fees, or detect life events such as a salary increase or a new child based on transaction clues to propose relevant services like investment plans or education savings accounts. AI algorithms can also run continuous A/B tests and learn which product offerings resonate with which customer profiles, constantly refining the personalisation strategy over time. This level of agility and insight is nearly impossible to achieve with manual analysis alone.

From a data perspective, effective personalisation relies on combining data from across the customer's relationship with the bank: core account data, mobile app interactions, credit history, and even external data like credit bureau scores. A unified data layer is again critical as it ensures that your AI tool has a 360-degree view of the customer to generate meaningful recommendations.



Make personalisation easy with Oradian's core and data layer

Oradian enables banks and lenders to easily leverage core banking data alongside other data streams to find these micro-segments and serve more relevant offers, such as tailored credit limits, installment plans, or product recommendations. The process can be fully automated: AI algorithms determine, say, which customers should get a notification about a new micro-insurance product, and the bank's system can send out personalised messages via SMS or in-app notifications. Over time, personalisation might evolve into autonomous financial planning for customers. For instance, imagine an AI that automatically moves a customer's money into optimal savings or investment options based on their goals, or that negotiates bills and finds the best loan deals on behalf of the customer.

One of the strategic advantages of AI-led personalisation is that it allows banks to deepen customer relationships without expanding headcount. A relatively small bank can effectively provide each customer with a personal experience through AI, something that traditionally only wealthy clients would receive via human advisors. This not only drives cross-sell and upsell, but also contributes to financial inclusion as customers receive guidance and product suggestions suited to their needs, which may help them use financial services more effectively through things like nudges to save, appropriate credit to build credit history, and insurance to protect assets.

To implement personalisation ethically, banks should ensure transparency and avoid overstepping. With robust data governance and a keen focus on customer benefit, AI-powered personalisation can deliver the kind of individualised attention that builds trust in a digital environment.

Automate the boring stuff

While customer-facing AI grabs headlines, an equally important application is using AI to streamline internal operations. Banks in emerging markets can greatly benefit from AI-driven efficiency gains, as they often operate with lean teams and must process high volumes of transactions or compliance checks with limited resources.

Operational AI covers a broad range of use cases: from automating repetitive back-office tasks like data entry, reconciliation, report generation to optimizing complex workflows like loan processing, customer onboarding, and case management in collections or compliance. The goal is to improve speed, accuracy, and scalability of processes that are critical to delivering services.



For example, instead of a loan officer manually reviewing and inputting financial statements for a credit application, an AI tool can automatically parse documents like tax returns, payslips, bank statements, and can then [pre-fill the relevant fields](#) in the loan origination system. What used to take hours or days can be done in minutes, allowing decisions to be made faster and staff to focus on analysis rather than data gathering. In fact, AI and ML have been shown to reduce processing times from days to minutes and cut the need for manual review by up to [60% in lending workflows](#), by automating data-intensive steps.

Another area where AI aids operations is workflow prioritisation and routing. In many banks, work queues for underwriting, servicing, or compliance can grow backlogged. AI algorithms can intelligently route tasks to the right employee or department and reprioritise work in real time based on factors like case urgency or customer value. For instance, in a collections scenario, AI could rank delinquent accounts by their likelihood of self-curing versus needing intervention, then prompt collectors to call the high-risk ones first. In customer onboarding, AI can scan document submissions and flag any missing or suspicious items before a human compliance officer starts the review, thereby preventing wasted time.

There are also examples of AI assisting in complex processes such as fraud investigations or AML by analysing large volumes of transaction data to highlight only the truly suspicious cases for human analysts and even drafting parts of investigative reports. In [Latin America](#), several banks now use AI-driven facial recognition and data matching to handle KYC/AML checks, which has accelerated customer onboarding while reducing errors and ensuring compliance with regulations. This kind of automation helps institutions meet regulatory requirements more reliably by eliminating manual lapses.

AI vs RPA: differences, similarities, and the benefits of each

RPA involves software robots mimicking the clicks and keystrokes of people to perform routine tasks across systems. This makes it particularly useful for integrating legacy systems or handling repetitive actions. AI, on the other hand, is great at making decisions or predictions based on data patterns.

In practice, the two often work together: AI figures out what should happen, while RPA executes how it happens in the system. For example, an AI model might review transactions and predict which ones are likely fraudulent and then an RPA bot can automatically lock those accounts or file a report in the compliance system. Both AI and RPA are far more effective when they operate on consistent, high-quality data rather than fragmented system-specific information.

This comes back to having that unified data backbone: if every bot or model uses the same source of truth data, the automation will be more reliable and easier to maintain. Oradian's platform supports this by offering open APIs and direct data access, enabling banks to connect their core system with various AI and automation tools seamlessly. In a case study from the Philippines, [Agribank integrated Oradian's core](#) with their loan origination and mobile apps via an open API layer, which allowed them to quickly roll out new automated workflows without rebuilding systems from scratch.



The benefits of AI and automation in operations manifest as lower operational costs, faster turnaround times, and fewer errors. Banks that embed AI in middle-office and back-office functions have seen dramatic reductions in manual workloads and improved operational resilience. One bank reported a 40% decrease in costs to verify customer identities thanks to AI-driven onboarding tools, highlighting how automation can melt away bottlenecks that previously required large amounts of staff.

Moreover, by automating routine tasks, banks can reallocate their human talent to higher-value activities like relationship management, complex problem-solving, and innovation. This shift is crucial for dynamic market institutions aiming to grow as this way, rather than simply hiring more people to handle growth, AI allows them to scale efficiently. However, it does require change management including training staff to work alongside AI, updating processes, and ensuring strong oversight of automated decisions.

When deploying AI in operations, banks should implement strategic designs for critical processes where a trained individual oversees or approves AI-driven actions, at least until the models have proven their accuracy. This balances efficiency with risk management and helps in gaining internal and regulatory trust in the AI systems. Ultimately, the successful use of AI in internal processes can make banks and lenders in dynamic markets more agile, cost-effective, and capable of handling growth, ultimately freeing them up to focus on strategic expansion and customer innovation.

40%

decrease in costs to verify customer identities thanks to AI-driven onboarding tools

Source: PwC (2025)



AI readiness checklist for banks

Use this checklist to assess how prepared your bank is to implement AI in a way that truly achieves value. It's designed for product, tech, data, risk, and compliance teams to complete together.

Data foundation

Do you have the data layer AI needs?

- ☒ We have a single, consolidated view of customers across products, channels, and devices.
- ☒ We can access detailed transactional and event data going back at least 12–24 months.
- ☒ Our data teams can query production-grade data off-core without risking downtime.
- ☒ We have basic data quality checks in place.
- ☒ We can bring in external or alternative data in a controlled way.

Ask yourself:

- When someone proposes an AI use case, can we say yes, we already have the data, or do we end up doing ad-hoc exports from core and Excel?

Core and integration readiness

Can your core and architecture support AI in production?

- ☒ Our core banking system exposes APIs that AI services or decision engines can call in real time.
- ☒ We can create a secure, read-only replica of core data for analytics and model training.
- ☒ We have a clear pattern for moving models from lab to live.
- ☒ We can add new decision logic into existing journeys without vendor tickets.
- ☒ Our monitoring tools can track how AI decisions impact latency, errors, and end-user experience.

Ask yourself:

- If we had a working model tomorrow, could we plug it into onboarding, lending, or fraud flows in weeks, or would we be blocked by the core?



Team and operating model

Do you have the people and processes to own AI, not just buy it?

- ☒ We have named owners for AI.
- ☒ Product, risk, and tech teams work together on experiments, not in silos.
- ☒ We have at least basic internal skills in data analysis, even if we still have to partner on advanced models.
- ☒ We can run small pilots with clear metrics without needing full projects.
- ☒ We have a culture of testing, learning, and iterating.

Ask yourself:

- If we started an AI pilot this quarter, do we know who would design it, who would own the metric, and who would decide if it scales?

Governance, risk and compliance

Can you use AI without ending up in regulatory or reputational trouble?

- ☒ We involve risk and compliance early in discussions about AI use cases.
- ☒ We document model assumptions, inputs, and limitations for any AI that affects customers.
- ☒ We can explain, at least in simple terms, why a customer was approved/declined or flagged by an AI system.
- ☒ We have basic monitoring in place for bias, drift, and error rates.
- ☒ There is a clear escalation path when AI systems fail or produce unexpected results.

Ask yourself:

- If a regulator walked in tomorrow and asked how AI is used in credit or fraud, could we show them a clear, defensible picture?

How to interpret your results

- **0–2 items** checked in any one section: You are not ready to rely on AI for real decisions. Your next step is to fix data access and integration patterns.
- **3–5 items** checked overall: Prioritise building a proper data layer off the core and a simple governance process.
- **15+ items** checked overall: You're in a good position to run focused pilots with measurable impact in 3–6 months.
- **Most boxes checked** across all four sections: You're AI-ready. Focus on selecting the right use cases, designing tight 90-day pilots, and building a repeatable AI delivery engine across teams.





How to implement AI

Adopting AI in banking is not a plug-and-play endeavor; it requires careful strategy, cross-functional coordination, and a phased approach.

Below are a few strategic guidelines for financial institutions in dynamic markets looking to implement AI effectively and responsibly.

Figure out what problem you're solving first

Identify specific, measurable problems that AI can solve, rather than pursuing AI for its own sake. Tie each AI initiative to a concrete KPI or business goal. For example, you might set a goal to reduce loan approval time by 20% or increase collections recovery rates by 15%. By defining success upfront and ensuring you have a way to measure it, you keep the project focused on delivering business value. If a metric isn't currently tracked, establish the measurement capability first. Clear objectives not only justify the investment but also rally your team around a common purpose.

Fix your data layer before doing anything else

As emphasised, data readiness is the foundation of AI success. Inventory your data assets and fix the basics: integrate siloed datasets, resolve inconsistencies, and eliminate poor-quality data. It's essential to set up a secure pipeline for data from your core systems into an analytics environment where AI models can be developed and tested. Many successful AI projects begin with establishing a data warehouse or enabling a core banking data replica like Oradian's Database Access to serve as a single source of truth.

Start small with a focused pilot project

Pick a use case that is impactful yet manageable in scope for your first AI project. Ideally, choose something where success can be demonstrated in a short timeframe like 3–6 months and where you have ample data. For instance, you might begin with an AI model to prioritise customer service inquiries, rather than a full-blown credit scoring overhaul.

Use historical data to build a simple, interpretable model first; like a logistic regression or decision tree for a risk scoring pilot. Avoid over-complicating with deep learning or black-box models initially as you need stakeholders and regulators to trust your results.



Involve key departments early, particularly Risk Management and Compliance, when designing the model and its inputs. This ensures the AI's decisions align with your institutional risk appetite and regulatory guidelines. The aim of a pilot is not to revolutionise your business overnight, but to create a proof of concept that delivers real value and learning. Keep the scope focused and success criteria clear.

Why you still need people making the decisions

Even as you automate decisions or processes with AI, keep team members in the loop, particularly in the early stages. Position AI as an augmented decision-making tool that can rank or recommend options, but people themselves should ultimately decide and approve, particularly for sensitive matters. This approach is wise for both internal acceptance and regulatory comfort in heavily supervised markets.

Establish an AI governance committee or working group that includes stakeholders from IT, data science, risk, compliance, and business units. Their job is to set guidelines on acceptable model uses, review AI outputs, and ensure proper controls (documentation, versioning, fallback procedures) are in place. Make transparency a priority: document how each model works, what data it uses, and how decisions are made. This not only helps with regulatory compliance but also builds trust internally. Many AI failures trace back to leaving Risk or Compliance out of the loop until late. Avoid this by having them co-own the project from the beginning. The governance body should also monitor outcomes once the AI is live: Are there any biases emerging? Are error rates within tolerance? Having a structured review process will help catch issues early and allow for course corrections.

Iterate in controlled pilots and scale up gradually

Once you build a model, test it in a controlled environment or on a subset of customers before full rollout. For example, if you develop an AI for loan underwriting, run it in parallel with human underwriters to compare decisions. Alternatively, introduce it in one branch or region first. Track pilot results rigorously: measure uplift against the old process and observe any unintended consequences.

When the pilot shows positive results and any issues have been resolved, integrate the AI solution into regular operations with proper change management. This might mean training staff on new workflows or updating policies. Even post-deployment, continue to monitor performance by setting a cadence to review whether model accuracy is holding up or if data drifts require retraining. By scaling gradually from pilot to broader implementation, you reduce risk and learn what works best in your specific context.



Your team isn't ready for AI (but they can be!)

AI adoption is as much about people as technology. Build internal capability by upskilling your team. You do this by training your analysts in data science basics, educating your business managers on what AI can and cannot do, and perhaps designating AI champions in each department.

Many institutions find they need to reskill a significant portion of their workforce, with one IBM study stating that [40% of the workforce needs to upskill](#) as a result of AI over the next three years, possibly translating into 1.4 billion people.

Encourage a culture of data-driven decision making and experimentation. This might involve bringing in a few data scientists or partnering with fintech specialists initially, but also plan for knowledge transfer to your internal team.

Secure executive buy-in and sponsorship for AI projects from the highest levels. When leadership actively supports AI and is educated on its benefits and risks, projects get the necessary resources and swift approvals. Executive champions can also help drive a culture that is open to innovation and not fearful of change. On the flip side, be transparent with employees about how AI will impact roles, demonstrating how it will augment their work, not replace them. Highlight success stories where AI took over drudgery and allowed employees to focus on more meaningful tasks. A positive, inclusive culture will smooth the adoption of AI and help attract the tech-savvy talent you may need.

Mistakes to avoid

Lastly, be wary of a few classic mistakes that have derailed AI initiatives in banking:

- Don't jump in without a solid data foundation. Fix data access and quality issues first, otherwise AI will amplify garbage-in, garbage-out problems.
- Don't let vendors dictate your strategy with shiny tools; start from your business goals and use technology as an enabler.
- Don't attempt overly ambitious moonshots initially; focus on achievable projects that deliver learning and ROI, then build on them.
- Don't exclude compliance/risk until late. Their early input can save you from hitting regulatory roadblocks at deployment.
- Don't expect AI to fix broken processes; if your loan process is fundamentally flawed, AI might highlight the issues but you'll still need to address the root causes.

Being mindful of these pitfalls will help ensure your AI journey actually yields sustainable improvements rather than becoming a costly experiment that goes nowhere.

By following these strategic guidelines, fintechs and banks can approach AI adoption in a practical, risk-managed way. Set the right goals, build the groundwork, and grow your AI capabilities step by step. Institutions that do this are far more likely to see measurable benefits within the first 12–18 months, such as faster turnaround times, reduced manual work, and improved risk outcomes, while positioning themselves for even greater AI-driven innovation in the future. Remember that AI is a journey, not a one-time project, and your strategy must evolve as the technology and regulatory landscape changes.



What your engineering team needs to know

Implementing AI in a banking environment involves several technical challenges and decisions. Below, we outline key considerations to ensure your AI projects are technically feasible, scalable, and secure.

How data gets from your core to your models

Again, a robust data architecture is non-negotiable. Banks should establish a data pipeline from the core banking system to an analytics environment. This often means leveraging APIs or database replication to continuously sync data. For example, a cloud-native core platform like Oradian offers live database replication with a secure, real-time copy of production data that can feed your BI dashboards, fraud detection systems, or ML models without impacting the performance of the live core.

Strive for one consolidated source of truth rather than ad-hoc data extracts. [Modern systems with an API-first design](#) greatly simplify integration as they allow AI engines, mobile apps, and other fintech services to plug into your core data and functionality with minimal custom code.

If you have a legacy core that's not as open, you may need to use middleware or RPA to pull data out, but this is less ideal long-term. Plan for a data warehouse or data lake that can store large volumes of structured and unstructured data, and implement extract-transform-load (ETL) processes to clean and prepare data for AI analysis.

On-premise vs cloud for AI workloads

AI workloads can be compute-intensive and spiky in demand. Cloud platforms allow you to scale processing power and storage up or down as needed without huge upfront investments in hardware. In dynamic markets, many banks have moved to cloud-based core systems like Oradian for this reason. Ensure your cloud setup or your SaaS core banking provider's setup has robust data security, encryption, and failsafes, as required by banking regulators.

Working in the cloud also eases integration of AI services, for instance, you can use cloud-based AI/ML services or libraries and connect them to your data via secure channels. Oradian's core banking system, for example, is hosted on secure AWS infrastructure with high availability and resilience, which supports data-driven operations and automation. That kind of environment is conducive to AI experimentation: you can make a sandbox, use real but masked data from your core, and test machine learning models without affecting production. Just be mindful of data residency rules as some countries require specific data to stay on local servers, which might influence your cloud strategy or require a hybrid approach.



Choosing your AI stack

Choose the right tools for your team's skill level and the task at hand.

This could range from off-the-shelf AI solutions like fraud detection services, chatbot platforms, or credit scoring engines to custom machine learning models built in-house.

If you lack a large data science team, starting with vendor solutions might be faster. However, ensure that any third-party AI tools can integrate with your systems via APIs or data feeds and that you can extract and store the results in your database for auditing. Many banks deploy a mix: for instance, a third-party chatbot platform integrated with their core via APIs, but an in-house model for credit risk that they built using historical loan data.

If building models internally, set up a proper ML Ops pipeline with version control for datasets and models, a process for deploying models into production, and monitoring to detect performance drift. It's wise to maintain a staging environment that mirrors production for testing new models or updates. Leverage your data replica or warehouse here so that you're testing on real, up-to-date data. Also, consider a feature store with a repository of commonly used data features that multiple AI models can use. For instance, Oradian's data backbone concept facilitates this by providing consistent core data for any model.

Black boxes won't fly with regulators

Always maintain documentation of model development: data used, variables selected, model parameters, and validation results. Implement controls to prevent unauthorised changes to models, this is akin to change management for software.

Many banks treat AI models under their existing model risk management framework, this involves periodic validation, stress testing, and sign-offs by a model risk committee. From a systems perspective, ensure you log all decisions the AI makes, for example, every time an AI model declines a loan or flags a transaction, record the event and the score/reason codes. This audit trail will help in both internal reviews and external audits. Technically, you might store these logs in a secure database or use your core banking system's logging capability. As regulation around

Build compliance in from day one

Any AI implementation must uphold the bank's security and privacy standards. When setting up data pipelines for AI, use encryption for data in transit and at rest. Limit access to sensitive data; your data science team might not need to see customer identities, for example, if they can work with anonymised IDs. Consider techniques like data masking or tokenization for building models on personal data and ensure that using customer data for AI is covered under your privacy policies and customer consents, particularly if you plan to use alternative data like phone records or social data. On the compliance front, imbue your AI systems with rules to enforce regulatory constraints.



AI can also be used to automate compliance, for instance, automatically flagging if a decision might violate fair lending rules. In terms of technical execution, incorporate checks or business rules engines that catch such scenarios before a decision is finalised. Additionally, perform thorough testing and validation for bias or disparate impact, especially for AI models that approve or deny credit or affect customer outcomes. This can be a technical process like analysing model output by demographic segments which again relies on having the data available.

Lastly, if an AI service crashes or produces obviously incorrect output, your operations should have a way to fall back to manual or rule-based processes. This might involve keeping the previous system running in parallel as a backup or being able to quickly disable the AI decision and route tasks to humans. Having such switches and contingency modes is part of good technical design in a critical industry like banking.

Working with vendors and other fintechs

Many emerging market banks partner with fintechs that specialise in AI solutions. When doing so, focus on integration capabilities:

- Does the partner provide APIs or SDKs?
- Can their system plug into your core or data warehouse easily?
- Will the data outputs from their AI flow back into your database?

Also consider latency, for example, for real-time use cases like fraud scoring each transaction, an on-premise or edge AI solution might be needed if internet connectivity is a constraint, whereas for less time-sensitive analytics, a cloud API call to a vendor could work. If you're working within an open banking ecosystem, leverage those APIs to gather richer data for your AI, remembering to get customer consent) For instance, using open banking data, you could feed transaction histories from accounts a customer holds at other banks into your own AI models for a more complete picture.

Ensure your technical team is equipped to manage multiple integrations and that you have proper API management and monitoring in place. Each integration point can be a failure point, so use robust error handling and fallback logic. The technical goal is to create a flexible architecture where you can slot in best-in-class AI components, whether they are built in-house or by partners, without having to reengineer your core systems each time. This is where a modern core like Oradian, with its modular and API-centric design, shows its value, as we deliver the flexibility, performance, and scalability required to modernise legacy infrastructure, enable streamlined integrations and automation-ready workflows all to banks and financial institutions in dynamic markets.



AI use case prioritisation matrix

Not every AI idea deserves immediate attention. This matrix helps you decide which AI projects to tackle first based on business impact, feasibility, and data readiness.

Step 1: List your candidate use cases

Examples include:

- AI credit scoring for thin-file borrowers
- Fraud anomaly detection
- Collections prioritisation
- Personalised offers in the app
- Chatbot assistant
- Onboarding risk scoring

Step 2: Score each use case

For each use case, score 1–3 on the dimensions below:

Business impact

- 1 = Nice to have
- 2 = Moderate impact (saves time or improves UX)
- 3 = High impact (revenue, losses, or regulatory priority)

Data readiness

- 1 = Data is scattered and incomplete
- 2 = Partial access; some cleanup needed
- 3 = Clean, accessible data already available

Technical feasibility

- 1 = Core changes are hard; vendor dependency high
- 2 = Some work can be done with APIs and moderate effort
- 3 = Work can be implemented using existing APIs, events, and known patterns

Time to value

- 1 = 12+ months
- 2 = 6-12 months
- 3 = 3-6 months

Total possible score per use case: 4-12

Step 3: Map use cases to priority levels

For each use case, score 1–3 on the dimensions below:

10-12 points: Tier 1

- High impact, high readiness.
- Great candidates for your first or second 90-day AI pilot.

7-9 points: Tier 2

- Valuable, but need data work or tech enablers.
- Use the next 1–2 quarters to fix data gaps or integration patterns.

4-6 points: Tier 3

- Either too speculative, too hard, or low impact for now.
- Keep them on a long list; don't spend scarce AI capacity here yet.

Step 4: Match priority to next actions

Tier 1: Pilot within the next 90 days

- Define the user journey and segment you'll run the pilot on.
- Use the 90-Day AI Pilot Planning Template to design the sprint.
- Make sure data access and integration patterns are locked in.

Tier 2: Get AI-ready over 3–12 months

- Improve data pipelines and quality for these use cases.
- Standardise logging, IDs, and events in the relevant journeys.
- Consider small analytical pilots to prepare for full AI deployment.

Tier 3: Reassess later

- Revisit annually or when strategy shifts.
- Some may get easier as your core, APIs, and data layer mature.



Prioritisation matrix template

CANDIDATE USE CASES	BUSINESS IMPACT	DATA READINESS	TECHNICAL FEASIBILITY	TIME TO VALUE	TOTAL SCORE	TIER
						Select
						Select
						Select
						Select
						Select
						Select
						Select
						Select

Business impact

- 1 = Nice to have
- 2 = Moderate impact (saves time or improves UX)
- 3 = High impact (revenue, losses, or regulatory priority)

Technical feasibility

- 1 = Core changes are hard; vendor dependency high
- 2 = Some work can be done with APIs and moderate effort
- 3 = Work can be implemented using existing APIs, events, and known patterns

Tiers

- Tier 1: 10-12 points
- Tier 2: 7-9 points
- Tier 3: 4-6 points

Data readiness

- 1 = Data is scattered and incomplete
- 2 = Partial access; some cleanup needed
- 3 = Clean, accessible data already available

Time to value

- 1 = 12+ months
- 2 = 6-12 months
- 3 = 3-6 months



Keep regulators happy with innovating

Banks in emerging markets must stay ahead of regulatory developments in AI to ensure that their AI initiatives meet compliance requirements and uphold customer trust. Regulators, for their part, are trying to balance innovation with risk mitigation; they see the promise of AI for financial inclusion and efficiency, but also worry about potential downsides like bias, lack of transparency, and data misuse.

Here are some of the key compliance considerations and challenges when deploying AI in banking, along with guidance on how to address them.

Get permission to use customer data

Many dynamic markets are still developing their data protection laws, which can be a double-edged sword: on the one hand, fewer restrictions might make it easier to use alternative data for AI, but on the other, a lack of clear regulations increases the risk of overstepping and damaging customer trust.

Banks and lenders should proactively adopt high standards for data privacy, even if not yet mandated by local laws. Ensure you obtain clear customer consent for using their data in AI-driven services, especially if you're leveraging data beyond the traditional banking scope, like phone records or social media data. Implement strict policies to prevent any use of data beyond what's consented and necessary.

A practical step is to anonymise or aggregate data whenever possible in the AI development process. Be as transparent as you can be with customers about the data you collect and how it benefits them, for instance, explain that analysing their utility payment history could help them get a loan they otherwise wouldn't qualify for. Even in jurisdictions without stringent privacy laws, global best practice, including aligning with principles of regulations like GDPR, will stand your institution in good stead as regulators catch up.

The bias problem no one wants to talk about

AI systems can inadvertently introduce or amplify biases present in historical data. This is a major concern for regulators and a serious reputational risk. For example, if historically a bank lent less to a certain region or demographic, a machine learning model might wrongly learn to be conservative with these specific demographic groups, thus perpetuating a bias. Emerging markets often have sensitive dimensions like ethnicity, gender, or rural vs urban divides that need careful handling to avoid discrimination.



Regulators are likely to rightly scrutinise AI-driven lending or decision systems for disparate impacts. To address this, build fairness checks into your model development. During testing, slice outcomes by protected attributes to ensure the AI's recommendations don't disproportionately disadvantage any one group. If they do, adjust the model and document both these results and these efforts. Some regulators may require explanations for how a model decision was fair and what factors it considered, be ready to provide that.

Explaining AI decisions to customers

Many countries are moving toward requiring that if a customer is adversely affected by an automated decision, like being denied a loan or flagged for investigation, they have the right to an explanation or a manual review. Even where it's not law, providing at least a minimal explanation is good practice.

Regulators will also expect transparency in terms of model documentation and auditability. This means maintaining detailed documentation of your AI models, as mentioned earlier, and being ready to present that to regulators if asked. Some markets might even introduce requirements to register certain AI algorithms with the central bank or undergo periodic audits. Staying ahead by self-auditing your AI and maintaining an audit trail can make compliance much smoother. Remember that trust is the currency of banking, being transparent about your use of AI will help avoid erosion of trust among customers and watchdogs.

Why one size won't fit all jurisdictions

Each country may have its own stance on AI, leading to a fragmented regulatory landscape for banks and lenders operating across borders. For instance, what data is acceptable for credit scoring or how e-KYC can be done with AI might differ by country.

If possible, participate in industry groups or sandbox programs to help shape emerging guidelines. When operating in multiple markets, be prepared to customise or geofence certain AI features. For example, a chatbot's functionality or an AI credit model might need tweaking if one country prohibits using certain data fields or requires a human sign-off for decisions. A practical approach is to design your AI systems to be configurable by market; have switches to turn off certain features or include additional checks to meet local regulations.



AI governance and risk frameworks

Regulators will expect banks to govern AI with the same rigor as other risks.

This means incorporating AI into your enterprise risk management.

Treat AI models as models under model risk management guidelines, including ensuring correct independent validation, back-testing, and ongoing performance monitoring. Ensure your board or senior management is informed about major AI initiatives and signs off on them, similar to approving a new credit policy.

Some regulators might require notification or approval for deploying certain AI systems, for instance, a central bank might want to approve any AI-based credit scoring system used by banks. Having an internal AI governance framework already in place can make these conversations easier. This framework should cover things like:

- How you select AI use cases
- How you validate data quality
- How you monitor for bias
- How you handle model updates
- How you train staff on AI oversight

In heavily regulated domains, a best practice is to use AI as a recommender rather than an autonomous decision-maker until both you and the regulator gain confidence.

Consumer protection and recourse

AI should not lead to consumer harm, whether through biased decisions, privacy invasion, or simply errors. It is important to have recourse mechanisms for customers: if an AI-powered process makes a mistake or a customer disagrees with an outcome, how can they get it corrected? For instance, if a customer is denied a loan by an AI system, ensure they can appeal the decision and have a person review it. Or if a chatbot fails to resolve an issue, have an easy way to talk to a live agent. Regulators might mandate such options, but even if not, they're good for customer satisfaction.

Monitor customer complaints for AI-related issues like “the chatbot didn’t understand me” or “I was unfairly denied service” as this can indicate where the AI needs improvement or where it’s creating risk of compliance issues.

With the growth of AI, we may also see new regulations around AI specifically, for example, requirements to disclose to customers when they are interacting with an AI vs a human, or rules on certain AI practices like web scraping data for credit scoring. Stay alert to such changes.

Ultimately, a regulator who sees a bank engaging in AI carefully and responsibly is more likely to green-light new experiments that can benefit everyone. Dynamic markets might have diverse rules, but a common thread is emerging: AI in finance must be done responsibly or not at all. Aligning with that ethos will position your institution as a leader as we head deeper into 2026.



Build an AI-ready future with Oradian

AI is set to be a defining force in banking through 2026 and beyond. For banks and lenders in dynamic markets, the potential rewards are immense; from more efficient operations and better risk management to richer customer experiences and expanded financial inclusion.

But as we have detailed, capturing these benefits starts with something more fundamental: a safe, stable way to run your core business and change it over time.

At the heart of it all is a simple truth: AI's power in banking is unlocked only when built on a strong foundation. It's something many businesses are getting wrong, with [95% of AI projects failing to deliver](#) on promises and poor quality data costing businesses over \$12 million annually and contributing to around 40% of failed business initiatives.

As a bank or lender, the foundation you need is a modern, bank-grade core banking infrastructure with robust, accessible data and governed change.

This is where Oradian comes in.

Oradian's cloud-native core banking platform is designed with the needs of high-growth, emerging market institutions in mind. It offers a secure, API-first architecture and real-time processing that make integrating AI solutions far easier. Oradian provides the missing layer that so many banks struggle with: direct, always-up-to-date access to your core data for analytics and AI purposes.

Through Database Access, Oradian gives clients a read-only replica of their production database, with full-fidelity data continuously synced and no impact on the live core's performance. In practice, this means your data scientists, analysts, or AI partners can plug into a clean, single source of truth, without having to build complex ETL pipelines or worry about core system load. It effectively solves the hardest part of AI integration with core banking systems: getting safe, reliable, consistently updated data into an environment where your teams can work. With Oradian, the lengthy and error-prone task of wrangling data is minimised, allowing you to focus on developing and deploying AI models that make a difference.



Already on Oradian? You may be closer to a working AI than you think

For institutions already using Oradian, you may be closer to AI-readiness than you think. Your core system is cloud-based and modular, which means adding new AI-driven features or connecting to fintech services can be done in weeks, not years.

Imagine kicking off an AI pilot today and within 90 days having a live solution, like an AI-powered collections prioritisation or a personalised offer engine, running in production and delivering measurable improvements. This is the kind of acceleration Database Access enables. It provides the confidence that whatever AI tool or model you choose, whether it's a credit scoring algorithm, a fraud detection engine, or a customer chatbot, it's operating on a solid, governed data foundation. You can swap in new models or vendors as needed, but your core data layer remains a steady backbone feeding every AI initiative.

The digital banking future is being written by those who are data-driven, agile, and bold enough to innovate responsibly. With a strong foundation beneath you, it's time to lead that future. Are you ready to embark on your AI journey and shape the future of finance in your market? With the right guideposts and the right platform, you can confidently say yes and we at Oradian are here to help make it happen.

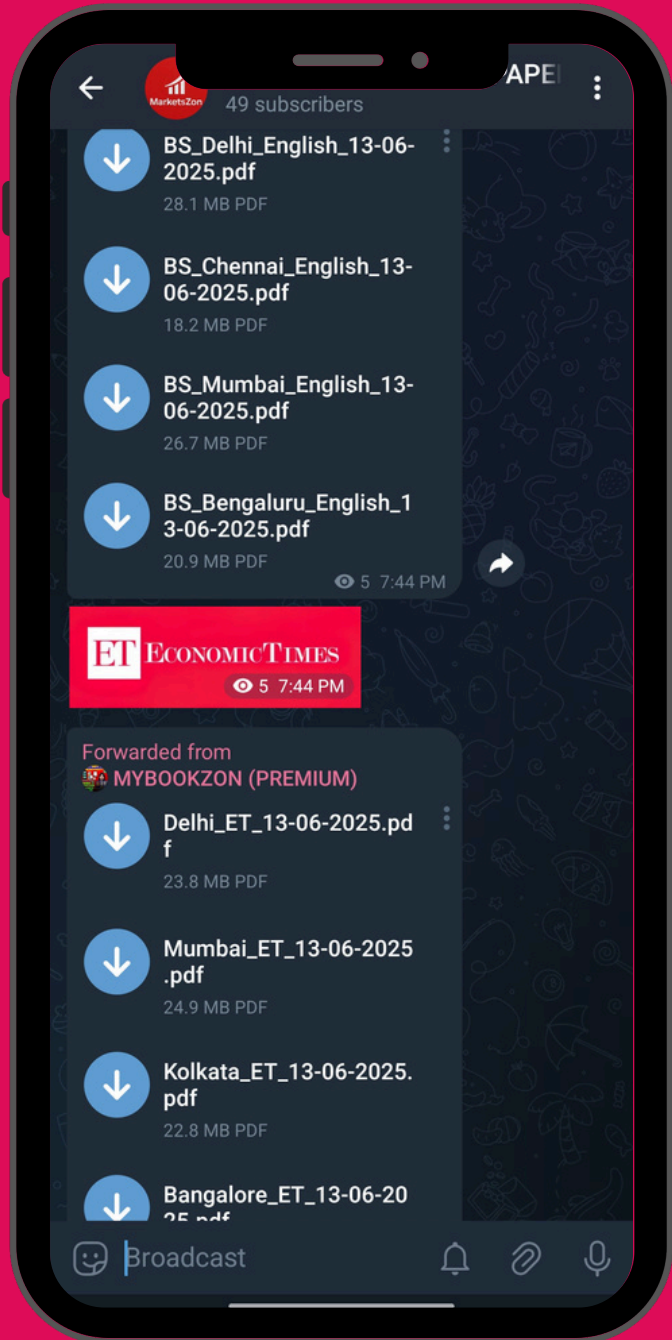
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